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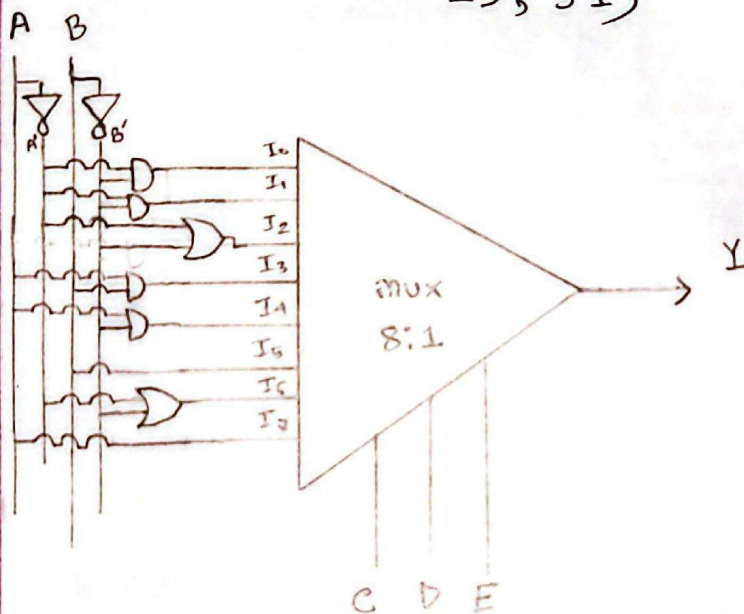
Section: 27A

Course Code: CSE260

[Assignment - 3]

[Ans to the question no 1]

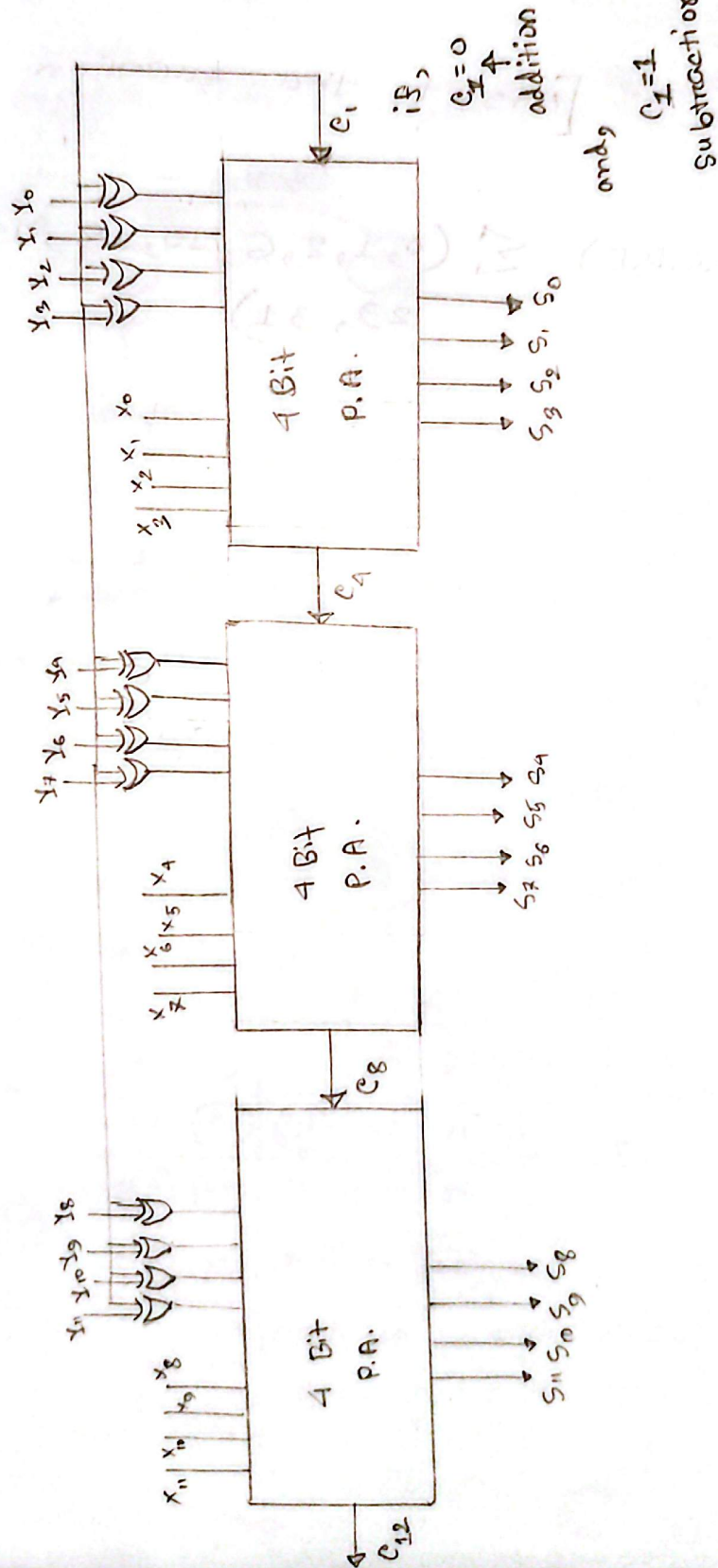
$$F(A, B, C, D, E) = \sum (0, 1, 2, 6, 10, 13, 14, 18, 20, 22, 23, 27, 29, 31)$$



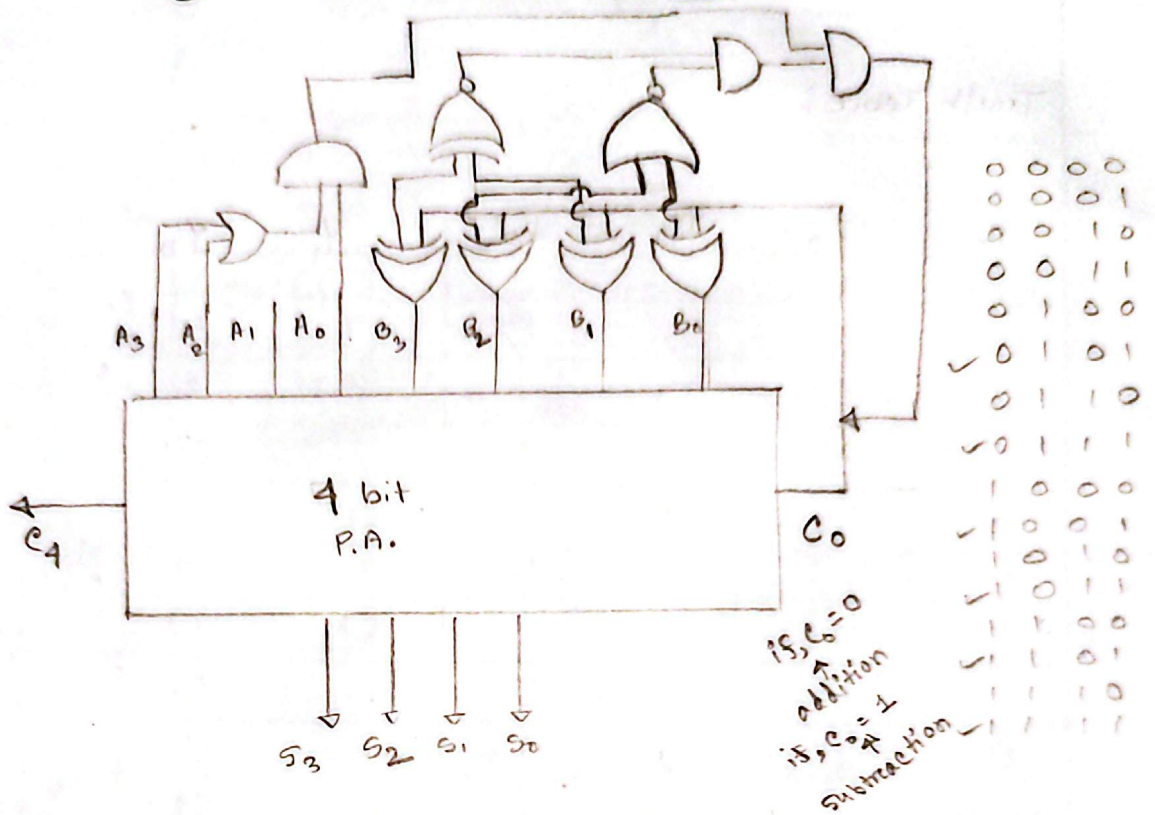
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$A'B'$	0	1	2	3	4	5	6	7
$A'B$	8	9	10	11	12	13	14	15
AB'	16	17	18	19	20	21	22	23
AB	24	25	26	27	28	29	30	31
	$A'B'$	$A'B$	$A'(B+B') + AB' = A' + AB' = A+B'$	AB	AB'	$B(A+A') = B$	$A' + AB' = A'+B$	$A(B+B') = A$

12-bit parallel adder-subtractor using 4-bit parallel adders

[Ans to the question no 2]



[Ans to the question no 3]



$A > 3$ and A is an odd number.

$$(B \% 10) == 0$$

$$(A_3 + A_2) A_0$$

$$(B_3 \odot B_1) (\overline{B_2 + B_0})$$

OR		NOR	
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

$$\begin{array}{cccc}
 A_3 & A_2 & A_1 & A_0 \\
 \hline
 0 & 1 & 0 & 1 \\
 0 & 1 & 1 & 1 \\
 1 & 0 & 0 & 1 \\
 1 & 0 & 1 & 1 \\
 1 & 1 & 0 & 1 \\
 1 & 1 & 1 & 1
 \end{array}$$

$B_3 B_2 B_1 B_0$

0	0	0	0	→	0
1	0	1	0	→	10

XNOR XNOR

[Ans to the question no 4]

Truth Table:

	I_3	I_2	I_1	I_0	Y_1	Y_0
	0	0	0	0	1	1
✓	0	0	0	1	0	1
✓	0	0	1	0	1	0
✓	0	0	1	1	0	1
✓	0	1	0	0	1	1
✓	0	1	0	1	0	0
✓	0	1	1	0	1	0
✓	0	1	1	1	1	1
✓	1	0	0	0	1	1
✓	1	0	0	1	1	1
✓	1	0	1	0	0	1
✓	1	0	1	1	1	0
✓	1	1	0	0	1	1
✓	1	1	0	1	0	1
✓	1	1	1	0	0	0
✓	1	1	1	1	x	x

For Y_1 ,

$I_3 I_2$ \ $I_1 I_0$	$I_1' I_0$	$I_1' I_0'$	$I_1 I_0$	$I_1 I_0'$
$I_3' I_2'$	1	0	0	1
$I_3' I_2$	1	0	1	1
$I_3 I_2$	1	0	X	0
$I_3 I_2'$	1	1	1	0

$$Y_1 = I_1' I_0' + I_3' I_0' + I_2 I_1 I_0 + I_3 I_2' I_0$$

For Y_0 ,

$I_3 I_2$ \ $I_1 I_0$	$I_1' I_0$	$I_1' I_0'$	$I_1 I_0$	$I_1 I_0'$
$I_3' I_2'$	1	1	1	0
$I_3' I_2$	1	0	1	0
$I_3 I_2$	1	1	X	0
$I_3 I_2'$	1	1	0	1

$$Y_2 = I_1' I_0' + I_2' I_1' + I_3 I_1' + I_3' I_1 I_0 + I_3 I_2' I_0'$$

